

Problem Set 3

ECON 2010 – Macroeconomics

Spring 2018

Due at the start of class Week 6, Wednesday, February 28. You have **one week** to complete this problem set. No late problem sets will be accepted. Please refer to the syllabus section on late assignments.

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct (some of) the problem set together in class.

Name:

Problem set instructions.

1. **Do not write on this problem set sheet** until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. Write your final answers to each question on a separate sheet of paper.
 3. Write your final answers in pen.
 4. Ensure that your name appears on this problem set and on the separate sheet(s) of paper with your answers, and staple your answers to the back of this problem set.
 5. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 6. Note that not all questions have *one* right answer. Some questions might have many correct answers.
-

Problem 1 (16 points)

For each of the statements below, do **each** of the following. **First**, indicate *which* component or components of GDP are related to the statement (your answer will be either consumption, private investment, government purchases, net exports, or none) (worth 1 point). **Second**, indicate how US GDP is affected (your answer will either be increases, decreases, stays the same, or unknown) (worth 2 points). **Third and finally**, indicate *by how much* US GDP is affected in a dollar amount (worth 1 point).

For each of the statements, use the principle of *all else equal* (a.k.a. ceteris parabis, holding other factors fixed) to conduct your analysis and answer the questions.

- (a) The city of Chattanooga purchases a new, German-made bus for \$150,000. The four tires on the bus were made in the US by Bridgestone tires in Tennessee. Each tire cost \$600. **(4 points)**
- (b) Aunt Bernice buys a new home in Rochester, NY in 2017 for \$600,000. The home was built in 2017 from \$40,000 worth of bricks produced in Canada in the same year. **(4 points)**
- (c) A digital camera assembly plant in Mexico buys \$100 worth of components from US factories, assembles a camera, and sells it to the Memphis Best Buy store for \$350. A Brazilian tourist buys it in Ann Arbor for \$400 while on her vacation, and takes it back with her to Brazil. **(4 points)**
- (d) The US government spends about \$80 billion in 2017 in new Medicaid payouts to US citizens and \$150 million in 2017 salaries of new Department of Health and Human Services employees to administer the expansion as a result of the Affordable Care Act (a.k.a. Obama Care). **(4 points)**

Problem 2 (28 points)

In this problem, consider the town of Springfield, USA. There are ten people who live in Springfield (and one dog, but let's not consider Santa's Little Helper for this problem). The table below was compiled by the US government and gives you the age and employment status of the ten residents.

The US government obtained information for column 3 ("Has a job?") by calling each of the residents of Springfield and asking them if they had a job. The information for column 4 ("Is looking for a job?") was obtained by asking each individual whether they were actively looking for a job. Slightly different from usual, the interviewer asked everyone whether they were looking for a job, whether they had a job or not.

Table 1: Age and employment status of the residents of Springfield, USA

Name	Age	Has a job?	Is looking for a job?
Maggie	1	No	No
Lisa	7	No	No
Bart	9	No	No
Homer	45	Yes	No
Marge	44	No	No
Mo	56	Yes	No
Mr. Burns	102	Yes	No
Barney	42	No	Yes
Grampa	79	No	No
Krusty the Clown	60	Yes	Yes

- (a) Using the information in the table and the definition of the unemployment rate, calculate the unemployment rate for Springfield. **(4 points)**
- (b) For each of the following changes, indicate whether the unemployment rate would *increase*, *decrease*, or *remain the same*. You do not need to calculate any change in the unemployment rate, but you may find it helpful to do so. Do not consider the changes cumulatively — that is, for each change, consider *only* that change relative to the information in the table. For each of the statements, use the principle of *all else equal* (a.k.a. *ceteris parabis*, holding other

factors fixed) to conduct your analysis. **(6 points total)**

(1) Lisa decides to start looking for work as a professor of astrophysics. (Yeah, she's that smart.) But she hasn't yet found a job. **(1 point)**

(2) Homer loses his job at the nuclear power plant after spilling jelly from a donut in his control panel and nearly causing a meltdown. He's perfectly happy taking salsa lessons and drinking at a local bar, so he doesn't look for additional work. **(1 point)**

(3) Krusty the Clown has been looking for a new job in the movie industry (he only does TV at the moment), but gives up on his search and stops looking for new work. **(1 point)**

(4) Marge decides to hire someone to clean the house and teaches her husband Homer how to cook dinner when he gets off work, and sets out to find a job as a hairdresser for blue haired women. She immediately finds and starts a hairdressing job. **(1 point)**

(5) Mr. Burns is officially declared dead when the US government determines that too many of his organs are plastic or animal parts. The government no longer considers him part of Springfield's population. **(1 point)**

(6) Both Grampa and Barney get part time jobs as checkers at the Kwik-E-Mart convenience store, but Barney keeps looking for full time work. **(1 point)**

(c) Using the information in the table and the definition of the labor market participation rate, calculate the labor market participation rate for Springfield. **(4 points)**

(d) For each of the following changes, indicate whether the labor force participation rate would *increase*, *decrease*, or *remain the same*. You do not need to calculate any change in the unemployment rate, but you may find it helpful to do so. Do not consider the changes cumulatively — that is, for each change, consider *only* that change relative to the information in the table. For each of the statements, use the principle of *all else equal* (a.k.a. ceteris parabis, holding other factors fixed) to conduct your analysis. **(6 points total)**

(1) Lisa decides to start looking for work as a professor of astrophysics. (Yeah, she's that smart.) But she hasn't yet found a job. **(1 point)**

(2) Homer loses his job at the nuclear power plant after spilling jelly from a donut in his control panel and nearly causing a meltdown. He's perfectly happy taking salsa lessons and drinking at a local bar, so he doesn't look for additional work. **(1 point)**

(3) Krusty the Clown has been looking for a new job in the movie industry (he only does TV at the moment), but gives up on his search and stops looking for new work. **(1 point)**

(4) Marge decides to hire someone to clean the house and teaches her husband Homer how to cook dinner when he gets off work, and sets out to find a job as a hairdresser for blue haired women. She immediately finds and starts a hairdressing job. **(1 point)**

(5) Mr. Burns is officially declared dead when the US government determines that too many of his organs are plastic or animal parts. The government no longer considers him part

of Springfield's population. **(1 point)**

(6) Both Grampa and Barney get part time jobs as checkers at the Kwik-E-Mart convenience store, but Barney keeps looking for full time work. **(1 point)**

(e) An old issue of the Springfield Inquirer newspaper from last year notes that the employment rate between two months dropped from 8% to 6%. The columnist claims that this is unambiguously good news for the Springfield economy. As a well-trained student of economics, you want to think about that claim. Give **two** reasons why the unemployment rate might have fallen. Write only one sentence (or statement) for each reason. Note that your reasons don't need to involve specific people or situations. You could consider your reasons normatively good or normatively bad — that isn't important. **(6 points total, 3 points each reason)**

(f) Write your own normative statement about the inclusion of those that are only employed part time in the calculation of the unemployment rate. Limit yourself to *one* sentence — *a* statement. You will not be graded on the normative content of your statement, but instead

whether you come up with a statement that is normative as opposed to positive. **(2 points)**

Problem 3 (11 points)

Consider three questions about an economy's production function.

- (a) Draw a generic production function that relates labor to production (output) and is shaped to reflect diminishing marginal returns to labor. By “generic” I mean that I’m not giving you enough information to plot specific points, and so it’s the shape of the curve that’s relevant. **(2 points)**

- (b) Draw a generic production function that relates labor to production (output) and is shaped to reflect increasing marginal returns to labor. By “generic” I mean that I’m not giving you enough information to plot specific points, and so it’s the shape of the curve that’s relevant. **(2 points)**

- (c) Re-draw the generic production function that you drew in part (a). Add a new curve to the plot with a dashed line that illustrates what would happen if the country whose production function you’ve drawn experienced a war that destroyed much of its infrastructure and machinery.

Consider no other changes in the economy except the war (i.e. use the principle of *ceteris paribus*). **(2 points)**

- (d) Re-draw the generic production function that you drew in part (a). Add a new curve to the plot with a dashed line that illustrates what would happen if the country whose production function you've drawn experienced an influx of low skilled workers (lower skilled than any workers currently in the country). **(2 points)**

- (e) Imagine two countries — let's say Canada and Cameroon. And let's simplify things and say that in each country there is only one farmer, and the only thing that the farmer produces is corn. There are only two inputs to growing corn — labor and land — and each are *equally* as productive in Canada and Cameroon. So, the one farmer, working the 1 acre of land in each country produces exactly the same amount of corn.

In Canada, the government taxes a small fraction of the farmer's corn production each year. That tax has always been small and always been the same. In Cameroon, the government sometimes taxes a small fraction of the farmer's corn, but sometimes takes all of it. It's unpredictable when this expropriation will occur, but it's happened a few times in the past few years.

Put yourself in the shoes of the farmer in each of these countries. The farmer in each country is trying to decide whether to make an investment in expensive fertilizer that will dramatically increase the amount of corn produced on a given acre of land. Given this information, in which country would you have a greater incentive to invest in the fertilizer? And in which country, therefore, would you expect GDP to be larger? Explain *briefly* why. **(3 points)**

Problem 4 (37 points)

While cleaning out old boxes from her attic, Enid finds some old receipts and documents from 1950. Use the information below to calculate different prices in different years.

Table 2: Approximate prices for a common basket of goods (U.S. Government data)

Cost of Basket in 1900	Cost of Basket in 1950	Cost of Basket in 2000
\$10.00	\$20.00	\$180.00

Table 3: 1950 prices in documents that Enid found

Item	Price
Enid's hourly wage	\$1.00
Hershey bar	\$0.05
One year of college tuition & room and board	\$1,200

(a) Given the information in the table, and 1950 as your base year, what is the CPI in each of the following years? **(6 points total)**

(1) 1900 **(2 points)**

(2) 1950 **(2 points)**

(3) 2000 **(2 points)**

(b) How much would Enid's 1950 hourly wage be worth in 2000? **(3 points)**

- (c) Enid's grandmother earned \$0.70 per hour as a store clerk in 1900. Is this wage in 2000 dollars more or less than what Enid's hourly wage is worth in 2000? (Hint: You may find it helpful to first convert Enid's grandmother's wage to 1950 dollars, and then convert the 1950 dollars to 2000 dollars. Although you don't have to. Be mindful of the numerator and denominator of the ratios that you're multiplying by the different prices. That is the trickiest part.) **(5 points)**
- (d) How much would Enid have needed to buy a Hershey's bar in 1900? **(3 points)**
- (e) How much should a Hershey's bar cost in 2000 dollars given the 1950 price and the CPI? **(3 points)**
- (f) Assume that there is no quality difference between a Hershey's bar in 2000 and a Hershey's

bar in 1950 (i.e. the bar is the same size, it has the same ingredients, same packaging, etc.). At Walmart in 2000, a Hershey bar cost \$1.20. Give a possible explanation for the difference between this price and the price you found in part (e) above. Please keep your answer **brief**. It's possible, for instance, to answer the question and get full credit in one sentence. **(5 points)**

- (g) Enid's granddaughter attends a private university where she pays \$50,000 per year in tuition, room and board. Is this more or less than the cost of tuition (in comparable prices, of course) that Enid found in her documents from 1950? That is, would \$50,000 in tuition have been more or less than what Enid paid in 1950 dollars in 1950? **(3 points)**

- (h) What was the inflation rate between 1900 and 1950 according to the table? **(3 points)**

- (i) What was the inflation rate between 1950 and 2000 according to the table? **(3 points)**

- (j) Consider two cities in 1950. In Jackson, the price of a gallon of milk is \$0.40. In Memphis, the price of an equivalent gallon of milk is \$0.36. Given only this information, would you conclude that inflation in Jackson is higher than Memphis, lower, the same, or unknown? **(3 points)**

Correcting Your Problem Set (8 points)

You'll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.